Risk Assessment - 3246RA

Information valid as of 20/05/2026 11:28

Low Risk

General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	02/07/2025
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	02/07/2026

Signed Off By	Oraib AlKheetanRojas
Signed Off On	19/05/2026
Sign Off Comments	

Operation Assessed	LiDAR Mapping Payload - Emesent Hovermap ST Operation

Associated with specific area

- Basement 2
- B2 029 (Robotics Lab)
- CTP

Description of work area and/or activity assessed

The Hovermap ST is a 360° spinning LiDAR payload used for autonomous 3-D mapping when handheld, robot-mounted, or carried by a drone. Typical activities include powering the unit from an external 12 – 54 V battery, initiating scans via tablet or Web UI, and downloading data after completion. Operation occurs mainly inside the Kinesis Lab and occasionally outdoors for field demonstrations.

Reason for Risk Assessment

To document and control health, safety, and equipment-damage risks introduced by routine use of the Hovermap payload during research and demonstrations.

Overall Current Risk

Low Risk

Average Number of Persons Affected

2	Employees
2	Students
2	Visitors

Hazards

Current Rating	Hazard Information	Task Status
Low Risk	**Electricity** Hovermap is powered through an external lead at 12 – 54 V DC (= 60 W). Accidental contact with damaged connectors or live conductors during installation, relocation, or battery swaps could cause mild electric shock, short-circuit, or arcing.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Power supply limited to SELV range and delivered through purpose-built, keyed cables. • “Power-on last / power-off first” rule enforced; the port remains de-energised until fully mated. • Visual inspection of leads and connectors before every use; damaged parts removed from service. • Only trained operators connect or disconnect power. • Battery charged and handled in accordance with manufacturer guidance; fire-rated Li-ion storage cabinet available.	

Potential Rating	Additional Controls Required
Low Risk	**No Remedial Actions Entered** If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk	**Moving Parts** The LiDAR sensor head spins continuously whenever a scan is active, and optional protective cages can also rotate with the payload. Fingers, hair, or loose clothing introduced into the path may be caught or struck; vibration may loosen insecure mounts.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Users keep hands clear of the rotating assembly; long hair tied back and loose clothing secured. • Protective cage or vehicle/pole mount used whenever Hovermap is handheld or lowered into confined spaces. • Spin-up and spin-down occur only after the unit is positioned away from personnel.	

Potential Rating	Additional Controls Required
Low Risk	**No Remedial Actions Entered** If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk	**Non Ionising Radiation** Hovermap employs multiple Class 1 eye-safe lasers for ranging. While the beams are enclosed, deliberate staring into an uncovered emitter or tampering with the optical assembly could present retinal hazard.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• The optical path is factory-sealed; no user-serviceable parts inside. Opening the enclosure is prohibited and voids calibration. • Operators instructed never to look directly into the sensor aperture at close range. • Routine cleaning performed with lens cloth only; solvents, abrasives, or mechanical scraping are forbidden.	

Potential Rating	Additional Controls Required
Low Risk	**No Remedial Actions Entered** If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.